

Textile Dyes



What is it?

Textile dyes are a large family of chemicals used to color fabrics. There are many different types.

Types of TEXTILE DYES:

Different types of textile dyes are used to dye specific fabrics:

- Acid dye: wool, silk, synthetic fiber, leather
- Azo dye: printing ink
- Basic dye: synthetic fiber
- Direct dye: cotton, leather, wool, silk, nylon
- Disperse dye: synthetic fiber
- Reactive dye: cotton, cellulosic fiber (the dye used in “tie dye” shirts)
- Sulfur dye: cotton, cellulosic fiber
- Vat dye: cotton, cellulosic fiber

Disperse dyes are the dyes that are most frequently found to cause allergy on patch testing. They are still frequently used to dye synthetic fibers, such as:

- Polyester
- Acetate and Triacetate
- Acetate Rayon
- Nylon
- Spandex/Lycra
- Acrylics and Modacrylics
- Polypropylene/Olefins
- Microfibers
- Blends that use the above fabrics or fibers

Disperse dye is not used in 100% cotton, rayon (viscose, viscose rayon), rayon variants (Modal, Tencel/Lyocell), silk, linen, wool, hemp, or bamboo.

Azo dyes are the most allergenic. They are rarely used in modern clothing for this reason.

What else should I know about TEXTILE DYE allergy?

Allergy to textile dyes often shows up as a rash at sites where friction or perspiration causes contact with the dye. Sweat may increase the release of dye from fabrics. Commonly affected areas are the inner thighs, backs of knees, buttocks, waist, genitalia, around the armpit and tops of feet (from dyes in shoe fabrics).

Avoidance will be specific to the dye to which you had a positive reaction on patch testing. It is important to keep in mind that most colors are created by blending dyes, therefore the causative clothing does not necessarily have to be the color to which you reacted on patch testing.

Most people will have fewer problems with light colored and loose-fitting garments. Another option is to remove offending dark linings and replace them with a white liner. Dyes are permanent and cannot be washed out of clothing. Repeated washing does not protect you from contact allergy to textile dyes. If a suspected garment must be worn, try 100% silk or cotton long underwear underneath it.

In addition, the ACDS Contact Allergen Management Program® (CAMP®) can help manage contact dermatitis. It helps to find products that are safe to use by creating a list of products that do not contain one's allergens. For more information about CAMP®, please ask your doctor or visit www.contactderm.org.

REVISION DATE: JANUARY 2026

DISCLAIMER: Every effort is made to ensure the accuracy of the information provided herein. However, The American Contact Dermatitis Society® makes no warranties or representations of any kind as to its accuracy or completeness. Such information is provided for informational purposes only and is not meant to be a substitute for professional medical advice, diagnosis, treatment, or care.